

Agentic AI for teaching

What we've been doing, what it costs, and where the judgment sits

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Today's plan



1:00 — Emily

- ▶ **Basics recap:** chatbots vs. agents, and what an agent can touch
- ▶ **Where AI can help w/ teaching prep:** story time
- ▶ **Grading and student data:** Principles and a pipeline suggestion



1:45 — Erkmen

- ▶ **What the evidence says** about AI in teaching so far
- ▶ **How others are using it**
- ▶ **Live demo:** an interactive dashboard your students can use

Chatbot vs. agent



A chatbot

- ▶ You type a message, it replies
- ▶ You copy the output somewhere
- ▶ It has no idea what's on your computer
- ▶ Each turn starts fresh



An agent

- ▶ **Reads your actual files** — slides, syllabus, problem sets, code
- ▶ **Runs commands** — Stata, Python, LaTeX, git
- ▶ **Remembers across turns** — standing instructions, course context
- ▶ Asks permission before doing anything risky

What it can touch — and how you widen it



1. Default

Your working directory only.

Asks before each action.
Where you start, and where
most work happens.



2. Widen

Grant more as trust grows.

More folders, standing permissions
for actions you've vetted
— loosened deliberately.



3. Sandbox / VM

Isolated environment.

For big, autonomous jobs — it
runs freely in a space that *can't*
touch the rest of your machine.

This matters for Part 2: **you decide what the agent can see.** Student data is the case where that decision has rules attached.

Part 1

Where it fits in teaching prep

What got cheaper, what didn't, and one story about Beamer

Two things it can buy you — 1. Less time

🕒 Less time

The obvious one: formatting, converting, “make this deck match that deck,” “turn these notes into a problem set.”

- ▶ The mechanical hour before class that used to be the whole prep
- ▶ Getting a draft of *anything* on the page so there's something to react to

Real, and it's why most people start — but on its own it mostly makes a *good-enough* lecture very, very cheap.



Two things it can buy you — 2. More range

More range

Things I used to *not do* because the fixed cost was too high in a busy semester:

- ▶ Bring in the paper I read last week
- ▶ A plug-and-play demo students can open in a browser
- ▶ One more worked example, at 11pm, the night before

It makes a **better** lecture **feasible**, not just a faster one.

A story about Beamer (2015)

Third time teaching statistics. Feeling like *quite* the expert educator. I decided to convert my ugly PowerPoint slides to Beamer, for reasons that seemed extremely important at the time.

Every week, on the couch: convert next week's lecture, slide by slide.

Not a great use of time on its face, but it was the right balance of tedious and engaging. I sat with every bullet point and asked what it was doing there.

Best semester I'd taught yet.

The only change students could see: no more PowerPoint.

 **It wasn't about the slides**

It was the **line-by-line pass** the silly task forced me into.

Complement, not substitute

Econometrics, this past year — what the workflow actually looked like:



The sequence

1. Write one giant prompt (Scott Cunningham's recommendation): standardize the format, add examples, tighten the arc
2. Let the agent work through the whole deck
3. **Then don't trust it.** Sit down and read the revised slides line by line, extremely slowly
4. ... which produced *more* changes than step 2 did — new activities, reordering, cuts



When it went well

A fresh pair of eyes on a deck: “Does this work? What's missing here?”
It made everything pretty *and* also a little bit broken



When I skipped step 3

It tried to sneak in a line that *not* violating the zero-conditional-mean assumption implies causality. 🕒

Same story as coding: the returns come from the verification pass, not from the generation.

Three workflows that worked — and two that didn't

✓ Worked

1. **Add new content.** Hand it a paper → draft slides in my format → I edit
2. **Audit a deck.** “Where are the holes? Does this hang together? What would a student get lost on?”
3. **Build a demo and embed it.** An interactive supply-and-demand explorer, a CLT visual — one HTML file, no login. (*Erkmen will show this.*)

If a workflow works, save it — as a prompt file or a skill — and reuse it next semester.

✗ Didn't

- ▶ Two hours arguing with Claude about the appearance of call-out boxes. (Students can't learn from ugly slides, apparently.)
- ▶ A JavaScript demo I was too excited about, and then lost my mouse in, mid-lecture

Every concern we have about students shortcutting their learning applies to us shortcutting our prep.

Choose your headwinds

If you want to set a personal best, you avoid the headwind. If you want to get *stronger*, a steady headwind is your friend — it keeps you pushing the whole way, no coasting.

Fast

If the goal is to go as fast as you can, these tools will do that — used well.

Better

If the goal is to be a better teacher, keep some friction — and **you get to pick which**.

Which frictions force the line-by-line pass? Which ones are just tedium? I don't have the general answer — but that's the design question.

Part 2

Grading and student data

The UVM rules in 30 seconds, and a rubric an agent can actually grade

The 30-second version (UVM edition)

FERPA applies the moment identifiable student work or grades go into an AI tool. Two clean paths.

✓ Path 1: UVM-approved tool

Microsoft Copilot, logged in with your NetID — approved for high-risk data including student academic records.

✓ Path 2: De-identify first

Strip the identifiers so it's no longer an education record. Then use whatever tool you like — including the good one you pay for yourself.

⊘ Not this path

Personal ChatGPT, personal Claude, Gemini: **not approved for UVM data**. (Yet!!) A named essay or a gradebook pasted into one is a disclosure the university hasn't authorized.

What counts as identifiable student data



FERPA covers...

- ▶ Grades, transcripts, class lists, student work
- ▶ Records **held by a vendor on your behalf** (they stay protected after handoff)
- ▶ **Direct** identifiers (name, ID, email)
- ▶ **Indirect** identifiers (DOB, writing style, topic)

The test: could a reasonable person in your community identify the student? If yes, it's identifiable.



The mosaic effect

In a **small seminar**, an “anonymized” essay *could* still be re-identifiable from:

- ▶ topic choice
- ▶ writing style
- ▶ facts you already know about the student

De-identification means addressing **combinations** — not just deleting the name.

A *student* pasting their *own* work into a consumer tool is their choice. The obligation runs to **you and the institution**.

Concrete classroom scenarios

Scenario	Verdict	Why
Named essay into personal ChatGPT for feedback	✗	Unauthorized disclosure
Same essay, fully de-identified , low re-ID risk, any tool	⚠	Defensible — your judgment
Same essay, in Copilot with your NetID	✓	UVM-approved for student data
Upload the class gradebook to summarize performance	✗	Grades are PII (unless Path 1)
Course chatbot trained on student submissions	✗	Needs approval + agreement
AI to draft a rubric, lecture, or problem set (no student data)	✓	No education records involved

The dividing line is almost always: **identifiable student data + a tool the university doesn't control**. Small seminars make “de-identified” harder than it sounds — topic and writing style re-identify.

A rubric an agent can grade



Five rules

1. **Cursory pass first.** Skim everything, list the edge cases, write the rules *before* grading (raw Stata output? null results? missing heading vs. missing section?).
2. **“Full points when...”** — criterion-based, not norm-referenced. “4 = exceptional” produces hedged 4.8s.
3. **Gates over sliders.** Word count, all questions present, attribution statement: binary and stable. Six-point scales drift.
4. **Split check from judge.** Checks (are there four peer-reviewed sources? is there an equation with the variables defined?) are where AI is reliable. Judgment stays yours.
5. **Plan the second pass.** Budget it. You’ll recalibrate.



“Literature” (4 pts), rewritten

```
gates (deterministic):
  >= 4 peer-reviewed in lit review
  distinct from the 2 intro sources
  NBER / World Bank / gov't = not

checks (agent - reliable):
  every source actually discussed
  each one linked back to the RQ
  what's known vs. still unknown

judge (you):
  synthesis, or summary?

score:  4 = gates pass + all checks
        3 = one check missed
        0 = gate fail -> flag
```

The original said “important literature discussed.” That’s a judgment word doing a gate’s job — and it’s the line every grader inflated.

What I asked them for

Research Paper: Research Proposal

Due by 1:15 PM on Thursday, April 9, 2026

Objective

The goal of this submission is for you to translate your research idea and data set into the outline of a workable paper. You can think of your research proposal as “baby paper”: a summary of what your question is, why it matters, and how you intend to solve it.

For this assignment, **some basic data work is necessary**. However, you may find it helpful to explore some analyses to get a better sense of what empirical specifications are feasible.

Components

While our first two assignments were fairly informal, this is a formal paper. That means that I will pay close attention to not only what ideas you present, but also *how* you present them.

Your proposal should be **at least 1200 words** (excluding references) and include the following components:

1. **Introduction** that contains (1) a clearly stated research question. What hypotheses are you testing? (2) Motivation – why is this important/interesting? Include at least 2 sources.
2. **Literature review**: discussion of related literature and how your paper fits in. Include at least 4 peer-reviewed sources that are *distinct from* your introduction sources (i.e., at least 6 unique sources total across the introduction and literature review).



What makes it gradeable

- ▶ **Countable things.** 1,200 words. 2 sources in the intro, 4 more in the lit review, 6 unique overall.
- ▶ **Named components.** Eight of them, each with its own rubric line — so “missing” is a fact, not an opinion.
- ▶ **38 points**, five levels per criterion, “full points when. . .” written out.

The assignment page *is* a grading input.
The agent reads this file before it reads a single submission.

What came in — except this one is fake

Paid Family Leave and Mothers' Labor Force Participation: Evidence from State Policies, 2010–2023

Andy Dwyer

ECON 3500 — Research Proposal — April 9, 2026

1. Introduction

How does paid family leave affect mothers? Since California introduced the first state paid family leave (PFL) program in 2004, eight states have adopted programs that replace part of a worker's wages for several weeks after the birth of a child. Supporters argue that paid leave keeps mothers attached to their jobs, while critics worry that it raises costs for employers and could reduce hiring of women of childbearing age. In this paper I ask whether state PFL laws increase the labor force participation of mothers with young children.

This question matters because women's labor force participation in the United States has stalled since the 1990s and has fallen behind most other rich countries. Blau and Kahn (2013) find that almost a third of the gap between the U.S. and other OECD countries can be explained by the absence of family-friendly policies such as paid leave. Goldin (2014) argues that the "last chapter" of gender convergence depends on the flexibility of jobs, and paid leave is one of the main policies that can change how costly it is for a mother to stay employed. The *family gap* in pay documented by Waldfogel (1998) also suggests that the years right after a birth are when women's careers are most vulnerable.

To answer this question I use the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) from 2010 to 2023, obtained through IPUMS. I compare mothers of children under five in states with a PFL program in effect to mothers in states without one, controlling for individual characteristics and for state and year fixed effects. This paper will show the causal effect of paid leave on mothers' employment and help policymakers in states that are currently debating PFL legislation.

2. Literature Review

The most studied PFL program is California's. Rossin-Slater, Ruhm, and Waldfogel (2013) use the CPS and a difference-in-differences design to show that California's program roughly doubled leave-taking among mothers of infants and increased weekly hours and wages of mothers of one- to three-year-olds. The effects were largest for less-advantaged mothers, who previously had little access to paid leave.

Baum and Ruhm (2016) use the National Longitudinal Survey of Youth 1997 and find that California's PFL raised the probability that mothers were working one year after birth by about

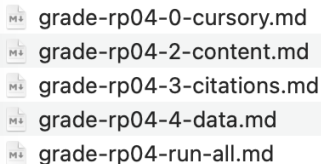


Why fake

- ▶ I can't put a real student's paper on this screen — so I wrote one. Fictional author, fictional numbers, real citations, a topic unlike anything my S26 class turned in.
- ▶ **13 flaws planted on purpose**, each mapped to a rubric line: a lit review that summarizes instead of synthesizes, a working paper counted as peer-reviewed, "this will *prove*" where it should hedge, a table with no group split, no clustering mentioned.
- ▶ **Answer key written first**, before any grader saw it. Target: **28.5/38** — deliberately mid, because that's where graders actually differ. (Real class: median 31.25.)

A grader you can't score is a grader you're trusting on vibes. This is the cheapest way I know to score one.

How it actually runs — four passes, one command



- grade-rp04-0-cursory.md
- grade-rp04-2-content.md
- grade-rp04-3-citations.md
- grade-rp04-4-data.md
- grade-rp04-run-all.md

>_ The fifth one is the whole trick

`run-all` doesn't grade. It launches the other four **in parallel**, then adds up each student's criterion scores and checks they match the reported total. I run one file.



One job each

- ▶ **cursory** — an overview pass. Skim all 21 first, list the edge cases, write the rules *before* anyone is graded.
- ▶ **content** — the assignment and the rubric. The 38 points, and the feedback the student actually reads.
- ▶ **citations** — a deliberately mechanical check: do these sources exist, are they peer-reviewed, do the counts hold? (*I care about this more than I can fully justify.*)
- ▶ **data** — will the data do what they say it will? Are the summary stats internally consistent? Does the specification match the data described?

Context I hand them: the assignment description, *and* my own feedback and grades on the two earlier assignments — which is what lets the feedback say “you did follow up on the concern I raised in February.”

For 21 submissions, I'm honestly not sure it was *faster*. It was more **consistent** — and that's the part I couldn't do by hand at 11pm.

Four graders, one proposal

A fake ECON 3500 proposal with 13 planted flaws. My real Spring 2026 grading prompt. Target: 28.5/38.

Component (pts)	Target	Copilot	Claude	Sonnet	Codex
Introduction (6)	4.5	6	6	6	6
Literature (4)	2	3	2	3	2
Data + table (6)	5	6	5	6	5
Empirical (10)	7	7.5	8	8	7.5
Threats (2)	1.5	2	2	2	2
Presentation (10)	8.5	9.5	9.5	9.5	9.5
Total (38)	28.5	34	32.5	34.5	32
Flaws deducted (of 13)	—	4	6	3	6
Flaws invented	—	0	0	0	0

Copilot with my NetID. Claude and Sonnet as fresh agents. Codex by hand.



What happened

- ▶ Everyone drifted up — on the judgment criteria.
- ▶ The checks held: both overclaims, the missing reference, the source count.
- ▶ Nobody hallucinated. Sonnet noticed nine flaws and deducted for three.

Read the notes, not the numbers.

What the feedback read like

Each grader also wrote the student feedback my prompt asks for. Compared to one I wrote in April.

Most useful

- ▶ **Codex.** My length. Handed the student the sentence to use instead of “prove.”
- ▶ **Claude.** Most complete. Three times my length.
- ▶ **Sonnet.** Fine. Called four summaries “genuinely good synthesis.”

Closest to my voice

- ▶ **Claude** got the structure: the name, the quoted sentences, the headers.
- ▶ **Codex** got the economy.
- ▶ **Nobody** got the pivot. No “That said,” no closing on the positive.

The substance is theirs. The voice is a five-minute edit.

What grading with agents actually taught me

✓ What worked

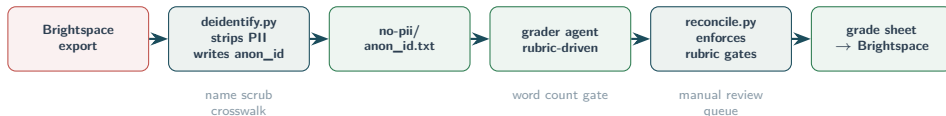
- ▶ **Different agents, different jobs.** A fair grader, a strict grader, an auditor, an injection scan — plus separate citation, data, and replication audits.
- ▶ **Checks beat scores.** 133 citations verified, 0 fabricated. The audits caught a log that crashed at the merge, a 14-line do-file, a blank template — content grading missed all three.
- ▶ **Disagreement is the product.** Only the papers where the graders disagreed came to me.

⚠ What to watch

- ▶ **Drift is up, and it's on the judgment lines.** You just saw it: four graders, four scores above the key, every gap on a criterion with an interpolated scale.
- ▶ **One pass is never enough.** Cursory → grade → audit → blind regrade → harshness audit. Hours to the first feedback document; more sessions to get it right.
- ▶ **Models differ — and it moves.** In April, Claude followed the rubric more faithfully than Codex; yesterday, on one proposal, Codex was closest. Check yours, and re-check.

Every correction I've had to make landed on a *judgment* criterion with an interpolated scale. None landed on a gate or a check.

What I do at scale — same rubric, different plumbing



At no point does the AI see a name — which is what lets me use whatever model follows the rubric best.

Why the plumbing

De-identify once → two graders (fair + strict) + an auditor → deterministic reconciliation → only disagreements reach me. The agent wrote the scripts; I described the rubric and the export.

When it's worth it

Research papers, 18 criteria, 17 students: yes. Three-question completion essays: Copilot and a good rubric are enough. The rubric is the same file either way.

“Should AI be grading at all?”

A fair question, and one a faculty attendee put to us bluntly at the June webinar. My honest answer, in three parts:

1. What I let it grade

Rubric-gated completion work — did they answer all six questions, at length, thoughtfully. Not the term paper. Not anything where the grade *is* the feedback.

2. It's an RA, not a referee

Same contract as a TA: it does a first pass, I check what it flags and spot-check what it didn't, and **the errors are mine either way**. Nothing about that changed.

3. Consistency is a process

My own first pass under-scored 11 of 17 papers; the second pass caught it. The comparison isn't AI vs. a perfect grader — it's AI-plus-review vs. me, essay 43 of 60, at 10pm.

What I don't have a settled answer on: **disclosure**. Do we tell students? What do we tell them? I'd genuinely like to know where this room lands.

Over to Erkmen

My half in one line: it makes a better lecture feasible and a good-enough one cheap · know the UVM rule · write a rubric with gates and checks · let the agent do the checks · you keep the judgment.

Erkmen: what the evidence says, how others are using it, and a live dashboard.